SURVEY | OPEN ACCESS

Efficient Transformers: A Survey

Authors: Yi Tay, Mostafa Dehghani, Dara Bahri, Donald Metzler | Authors Info & Claims

ACM Computing Surveys, Volume 55, Issue 6 • Article No.: 109, Pages 1 - 28 • https://doi.org/10.1145/3530811

Published: 07 December 2022 Publication History

Abstract

AI Summary

Transformer model architectures have garnered immense interest lately due to their effectiveness across a range of domains like language, vision, and reinforcement learning. In the field of natural language processing for example, Transformers have become an indispensable staple in the modern deep learning stack. Recently, a dizzying number of *"X-former"* models have been proposed—Reformer, Linformer, Performer, Longformer, to name a few—which improve upon the original Transformer architecture, many of which make improvements around computational and memory *efficiency*. With the aim of helping